

# Cloud Computing Practical Examination

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## Instructions

Perform all tasks using your Linux Virtual Machine.

Capture screenshots wherever required.

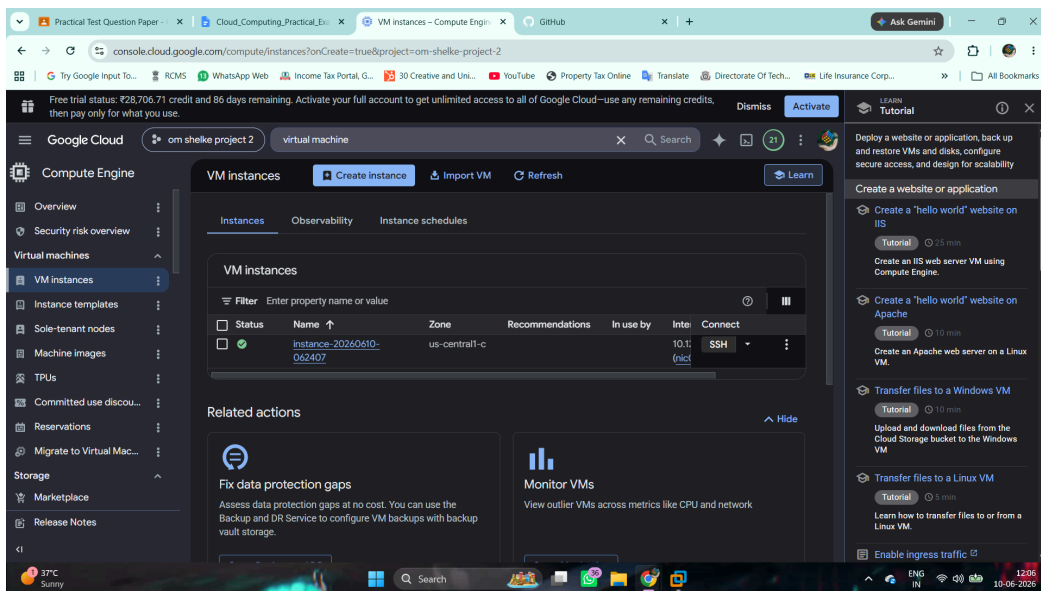
Store all shell scripts inside a folder named practical\_exam.

Push all scripts and supporting files to a GitHub repository.

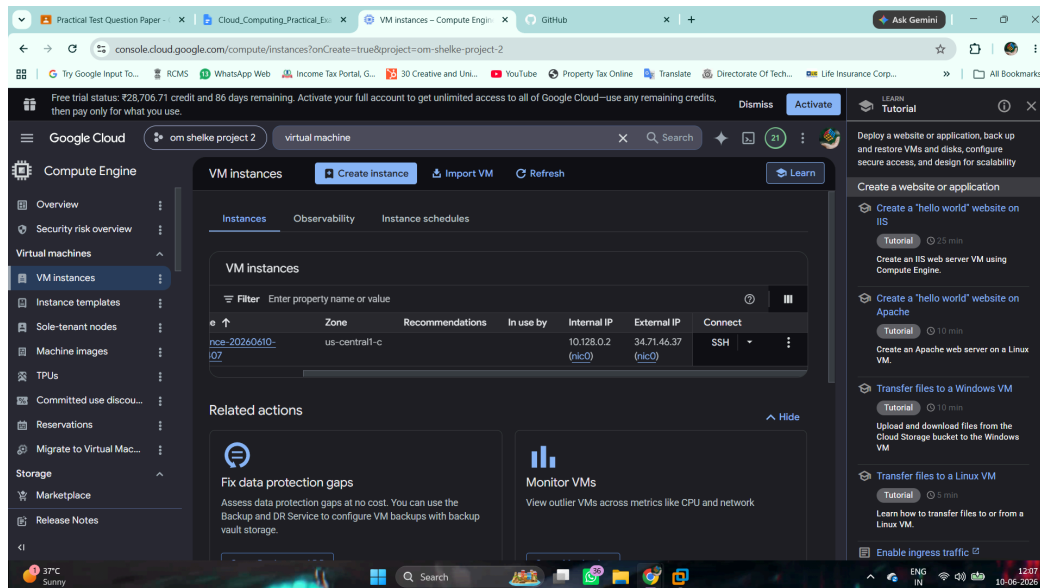
Submit a PDF report containing screenshots, outputs and GitHub repository URL.

## Part A – Cloud & Linux VM Setup

Task 1: Create a Linux Virtual Machine with Public IP and SSH access enabled.



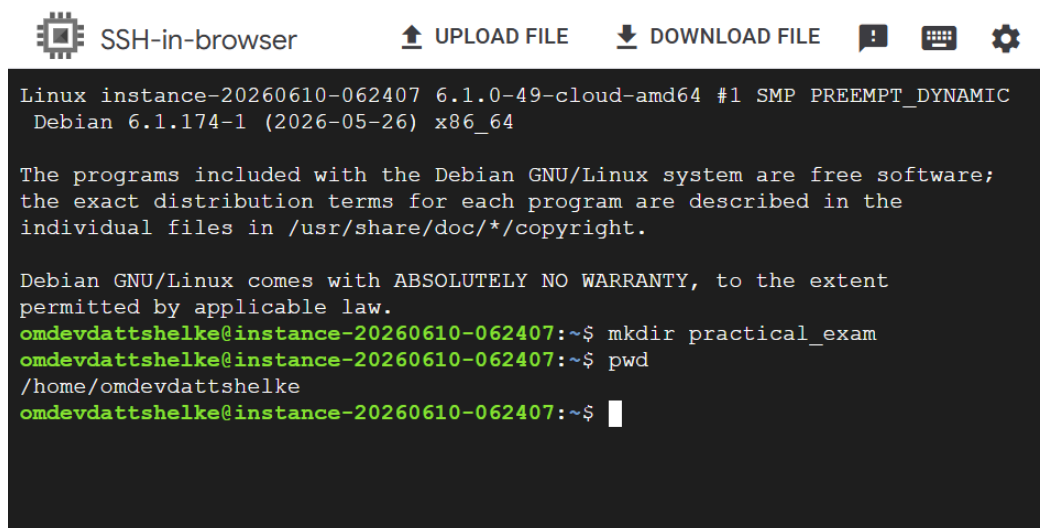
Task 2: Connect to the VM using SSH and capture screenshots.



Task 3: Create practical\_exam/scripts, practical\_exam/outputs and practical\_exam/logs.

## Part B – Linux Commands

Task 4: Display current working directory, current user and hostname. Save output.



Task 5: Create 5 files inside scripts folder.



SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE



```
omdevdattshelke@instance-20260610-062407:~$ cd practical_exam
omdevdattshelke@instance-20260610-062407:~/practical_exam$ mkdir scripts
omdevdattshelke@instance-20260610-062407:~/practical_exam$ cd..
-bash: cd..: command not found
omdevdattshelke@instance-20260610-062407:~/practical_exam$ cd ..
omdevdattshelke@instance-20260610-062407:~$ mkdir practical_output
omdevdattshelke@instance-20260610-062407:~$ cd practical_exam
omdevdattshelke@instance-20260610-062407:~/practical_exam$ dir
scripts
omdevdattshelke@instance-20260610-062407:~/practical_exam$ cd scripts
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ touch o
ne.txt two.txt three.txt four.txt five.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ ls
five.txt four.txt one.txt three.txt two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$
```

Task 6: Copy two files into outputs folder.



SSH-in-browser

UPLOAD FILE

DOWNLOAD FILE



```
omdevdattshelke@instance-20260610-062407:~$ cd practical_exam
omdevdattshelke@instance-20260610-062407:~/practical_exam$ mkdir scripts
omdevdattshelke@instance-20260610-062407:~/practical_exam$ cd..
-bash: cd..: command not found
omdevdattshelke@instance-20260610-062407:~/practical_exam$ cd ..
omdevdattshelke@instance-20260610-062407:~$ mkdir practical_output
omdevdattshelke@instance-20260610-062407:~$ cd practical_exam
omdevdattshelke@instance-20260610-062407:~/practical_exam$ dir
scripts
omdevdattshelke@instance-20260610-062407:~/practical_exam$ cd scripts
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ touch one.txt
two.txt three.txt four.txt five.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ ls
five.txt four.txt one.txt three.txt two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ cp one.txt tw
o.txt output
cp: target 'output': No such file or directory
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ mkdir output
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ cp one.txt tw
o.txt output
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ ls
five.txt four.txt one.txt output three.txt two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts$ cd output
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ ls
one.txt two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 7: Rename one file.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ mv one
.txt six.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ ls
six.txt two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 8: Delete one file.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ rm six
.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ ls
two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 9: Display hidden files.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ ls -a
.  ..  two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 10: Display disk usage information.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ ls -a
.  ..  two.txt
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G   0    1.9G   0% /dev
tmpfs           393M  536K  392M   1% /run
/dev/sda1       9.7G  2.7G  6.6G  29% /
tmpfs           2.0G   0    2.0G   0% /dev/shm
tmpfs           5.0M   0    5.0M   0% /run/lock
/dev/sda15     124M  12M   113M  10% /boot/efi
tmpfs          393M   0   393M   0% /run/user/1000
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 11: Display memory usage information.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
f
ree -h

              total            used            free           shared  buff/cache   ava
ilable
Mem:          3.8Gi          532Mi          2.4Gi           536Ki        1.1Gi
3.3Gi
Swap:          0B             0B             0B
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 12: Display running processes.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
ps
  PID TTY          TIME CMD
 1597 pts/0    00:00:00 bash
 3233 pts/0    00:00:00 ps
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.1  0.3 167976 12544 ?        Ss   06:33   0:02 /sbin/i
root         2   0.0  0.0      0     0 ?        S    06:33   0:00 [kthrea
root         3   0.0  0.0      0     0 ?        I<   06:33   0:00 [rcu_gp
```

Task 13: Find all .sh files inside practical\_exam.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
find . -type f -name "*.sh"  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

## Part C – Permissions & Users

Task 14: Create secure\_data.txt and apply permission 640.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
touch secure_data.txt  
chmod 640 secure_data.txt  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 15: Demonstrate Owner, Group and Others using commands.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
ls -l secure_data.txt  
-rw-r----- 1 omdevdattshelke omdevdattshelke 0 Jun 10 07:02 secure_data.tx  
t  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 16: Display User ID, Group ID and assigned groups.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
id  
uid=1000(omdevdattshelke) gid=1003(omdevdattshelke) groups=1003(omdevdatts  
helke), 4(adm), 30(dip), 44(video), 46(plugdev), 1000(google-sudoers), 1001(dock  
er), 1002(lxd)  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

## Part D – Shell Scripting

Task 17: Script to print Welcome to Cloud Computing.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
echo "Welcome to Cloud Computing"  
Welcome to Cloud Computing  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 18: Script accepting username and displaying Welcome .

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
read -p "Enter your username: " username  
echo "Welcome, $username!"  
Enter your username: om  
Welcome, om!  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 19: Script to find larger of two numbers using if-else.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
read -p "Enter first number: " num1  
read -p "Enter second number: " num2  
  
if [ $num1 -gt $num2 ]; then  
    echo "$num1 is larger than $num2."  
elif [ $num2 -gt $num1 ]; then  
    echo "$num2 is larger than $num1."  
else  
    echo "Both numbers are equal."  
fi  
Enter first number: 10  
Enter second number: 20  
20 is larger than 10.  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 20: Script to determine even or odd number.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
read -p "Enter a number: " num  
  
if [ $((num % 2)) -eq 0 ]; then  
    echo "$num is Even."  
else  
    echo "$num is Odd."  
fi  
Enter a number: 10  
10 is Even.  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 21: For loop displaying 1 to 20.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
for i in {1..20}  
do  
    echo $i  
done  
1  
2  
3  
4  
5  
6  
7  
8  
9  
10  
11  
12  
13  
14  
15  
16  
17  
18  
19  
20  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 22: While loop displaying 20 to 1.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
count=20  
while [ $count -ge 1 ]  
do  
    echo $count  
    ((count--))  
done  
20  
19  
18  
17  
16  
15  
14  
13  
12  
11  
10  
9  
8  
7  
6  
5  
4  
3  
2  
1  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
```

Task 23: Multiplication table script.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
read -p "Enter a number for the multiplication table: " num  
  
for i in {1..10}  
do  
    echo "$num x $i = $((num * i))"  
done  
Enter a number for the multiplication table: 50  
50 x 1 = 50  
50 x 2 = 100  
50 x 3 = 150  
50 x 4 = 200  
50 x 5 = 250  
50 x 6 = 300  
50 x 7 = 350  
50 x 8 = 400  
50 x 9 = 450  
50 x 10 = 500  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 24: Count files inside scripts directory.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
#!/bin/bash  
# Lists all filenames ending in .sh in the current directory  
echo "Shell script files in the current directory:"  
ls *.sh 2>/dev/null || echo "No .sh files found."  
Shell script files in the current directory:
```

Task 25: Display all shell script filenames.

Task 26: Display current date and time and save output.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
date  
Wed Jun 10 07:13:42 UTC 2026  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

## Part E – Processes & Scheduling

Task 27: Save running processes to outputs/processes.txt.

```
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
# Ensure the outputs directory exists  
mkdir -p outputs  
  
# Save the running processes  
ps aux > outputs/processes.txt  
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$  
█
```

Task 28: Identify top 5 memory-consuming processes.



```

omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$
ps aux --sort=-%mem | head -n 6
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root          1442  1.3   3.9 2073180 158912 ?        Ssl  06:33    0:34 /opt/go
ogle-cloud-ops-agent/subagents/opentelemetry-collector/otelopscol --config
=/run/google-cloud-ops-agent-opentelemetry-collector/otel.yaml
root          1440  0.0   1.5 1868812 64188 ?        Ssl  06:33    0:00 /opt/go
ogle-cloud-ops-agent/libexec/google_cloud_ops_agent_wrapper -config_path /
etc/google-cloud-ops-agent/config.yaml -log_path /var/log/google-cloud-ops
-agent/subagents/logging-module.log /opt/google-cloud-ops-agent/subagents/
fluent-bit/bin/fluent-bit --config /run/google-cloud-ops-agent-fluent-bit/
fluent_bit_main.conf --parser /run/google-cloud-ops-agent-fluent-bit/fluen
t_bit_parser.conf --storage_path /var/lib/google-cloud-ops-agent/fluent-bi
t/buffers
root           388  0.0   1.5 1288792 60916 ?        Ssl  06:33    0:01 /usr/bi
n/google_osconfig_agent
root          1134  0.0   0.6 1279572 27928 ?        Sl   06:33    0:01 /usr/li
b/google/guest_agent/GuestAgentCorePlugin/core_plugin --protocol=unix --ad
dress=/run/google-guest-agent/plugin-connections/GuestAgentCorePlugin_2026
0511.00.sock --errorlogfile=/var/lib/google-guest-agent/481642438515207268
7/agent_state/plugins/GuestAgentCorePlugin/plugin.log
root           1013  0.0   0.6 1278188 26924 ?        Ssl  06:33    0:02 /usr/bi
n/google_guest_agent_manager
omdevdattshelke@instance-20260610-062407:~/practical_exam/scripts/output$

```

Task 29: Create a cron job appending current date and time every minute.

```
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
```

Task 30: Verify cron execution with evidence.

```
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
~
~
~
~
~
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~
~
~
```

## **Part F – Git & Submission**

Task 31: Create GitHub repository cloud-practical-exam.

Task 32: Initialize Git in practical\_exam directory.

Task 33: Add all files.

Task 34: Commit changes with message Cloud Practical Exam Submission.

Task 35: Push files to GitHub.

Task 36: Verify repository contains scripts, outputs, screenshots and documentation.

## **Final Deliverables**

PDF report containing screenshots and outputs.

GitHub Repository URL.

All shell scripts.

Evidence of cron execution.

Evidence of successful SSH connection.